QUBIT ZERO TO HERO

Complex numbers and phase

S01 · Mathematical foundations

Cartesian form, magnitude and phase
Step-by-step calculations and Python checks

The imaginary unit

$i^2 = -1$

A real square cannot equal −1. Complex numbers extend the real numbers.

In mathematics we write i. In Python we write 1j.

Example: 3 + 4i in mathematics is 3 + 4j in Python.

Cartesian form and arithmetic

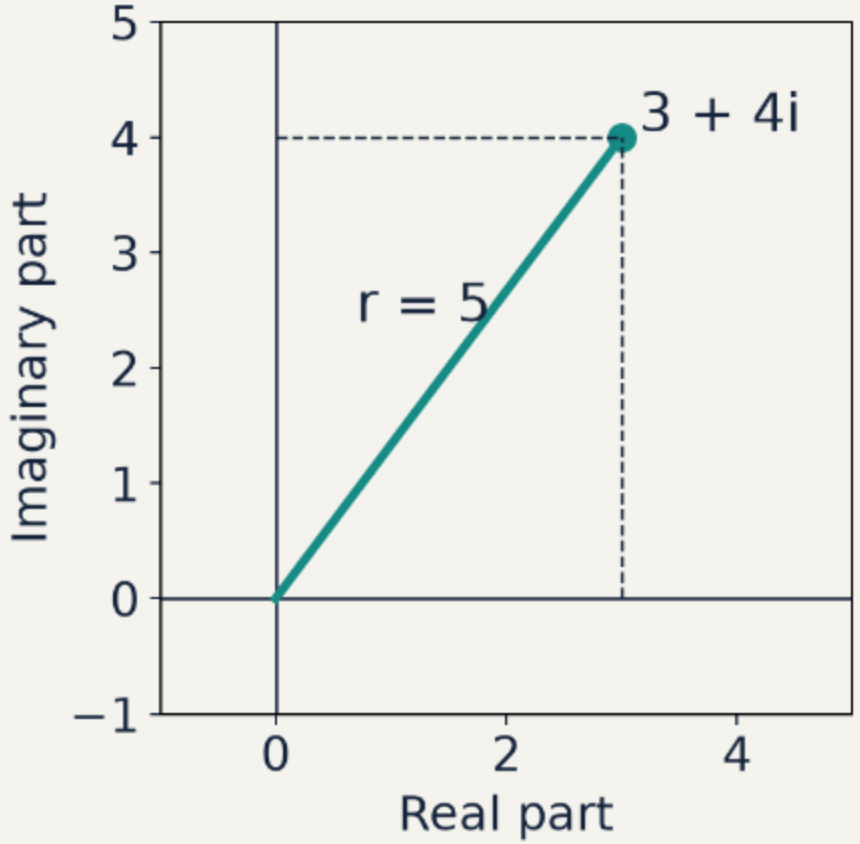
$z = a + ib$

$a = \mathrm{Re}(z)$ and $b = \mathrm{Im}(z)$, with a and b real.

$$(a + ib) + (c + id) = (a + c) + i(b + d)$$

$$(a + ib)(c + id) = (ac - bd) + i(ad + bc)$$

The complex plane



3 + 4i is the point (3, 4).

Horizontal axis: real part

Vertical axis: imaginary part

Complex conjugation

$z^* = a - ib$

Conjugation flips the sign of the imaginary coordinate.

If $z = 3 + 4i$, then $z^* = 3 - 4i$.

Useful check: $(z^*)^* = z$.

Squared magnitude, line by line

$$
\begin{aligned}
z^* \times z &= (a - ib)(a + ib) \\
&= a^2 + iab - iab - i^2 b^2 \\
&= a^2 + b^2 = |z|^2
\end{aligned}
$$

The cross terms cancel. Since $i^2 = -1$, the last term is $+b^2$.

The result is real and nonnegative.

Amplitude and probability

$$p_0 = |\alpha|^2 \qquad p_1 = |\beta|^2$$
$$|\alpha|^2 + |\beta|^2 = 1$$

A normalized qubit state has amplitudes α and β for outcomes 0 and 1.

An arbitrary complex number need not be a valid normalized amplitude.

Example: $|3 + 4i|^2 = 25$, so this alone cannot be a probability.

A complex square is different

$$z = (1 + i)/2$$
$$z^2 = i/2$$
$$|z|^2 = (1^2 + 1^2)/4 = 1/2$$

z^2 multiplies z by itself. $|z|^2$ multiplies z^* by z.

The probability contribution is 1/2, a real number.

Polar coordinates and radians

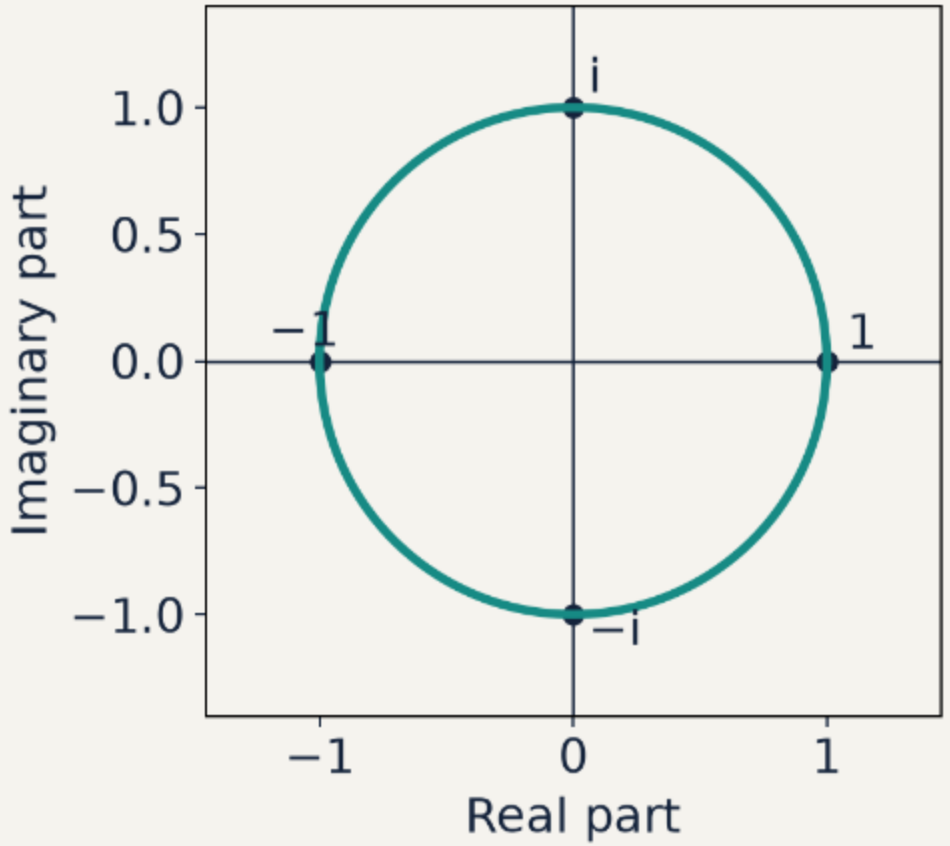
$$z = r(\cos \varphi + i \sin \varphi)$$
$$r = |z| \qquad \varphi = \operatorname{atan2}(b, a)$$

r is the distance from the origin. φ is the phase angle.

π radians = 180°. A full turn is 2π radians.

The phase of zero is undefined. atan2 preserves the quadrant.

Unit magnitude and phase



r = 1 on the unit circle.

φ = 0 gives 1.

φ = π/2 gives i.

φ = π gives −1.

Euler’s identity and polar form

$$e^{i\varphi} = \cos\varphi + i\sin\varphi$$

$$z = r\,e^{i\varphi}$$

$$|e^{i\varphi}|^2 = \cos^2\varphi + \sin^2\varphi = 1$$

The exponential is a compact way to write the same complex number.

At $\varphi = \pi$: $e^{i\pi} = -1$, hence $e^{i\pi} + 1 = 0$.

Multiplication adds phase

$$z_1 z_2 = r_1 r_2 \, e^{i(\varphi_1 + \varphi_2)}$$
$$(r\, e^{i\varphi})^* = r\, e^{-i\varphi}$$

Multiplication multiplies magnitudes and adds phases.

Conjugation reverses phase while preserving magnitude.

Example: $(2i)(-i) = 2$, with phase $\pi/2 - \pi/2 = 0$.

Example 1 · Cartesian to polar

$$z = 3 + 4i$$

$$r = \sqrt{(3^2 + 4^2)} = 5$$

$$\varphi = \operatorname{atan2}(4, 3) \approx 0.927295 \text{ rad}$$

$$z \approx 5\, e^{(i \times 0.927295)}$$

The angle is about 53.130°. The exponential uses radians.

Example 2 · A probability contribution

$$\alpha = (1 + i)/2$$
$$\alpha^* = (1 - i)/2$$
$$|\alpha|^2 = \alpha^*\alpha = (1 + 1)/4 = 1/2$$
$$|\alpha| = 1/\sqrt{2}$$

A second amplitude with squared magnitude 1/2 completes normalization.

Example 3 · Phase rotation

$$\alpha = (1 + i)/2 \qquad \varphi = \pi/2$$

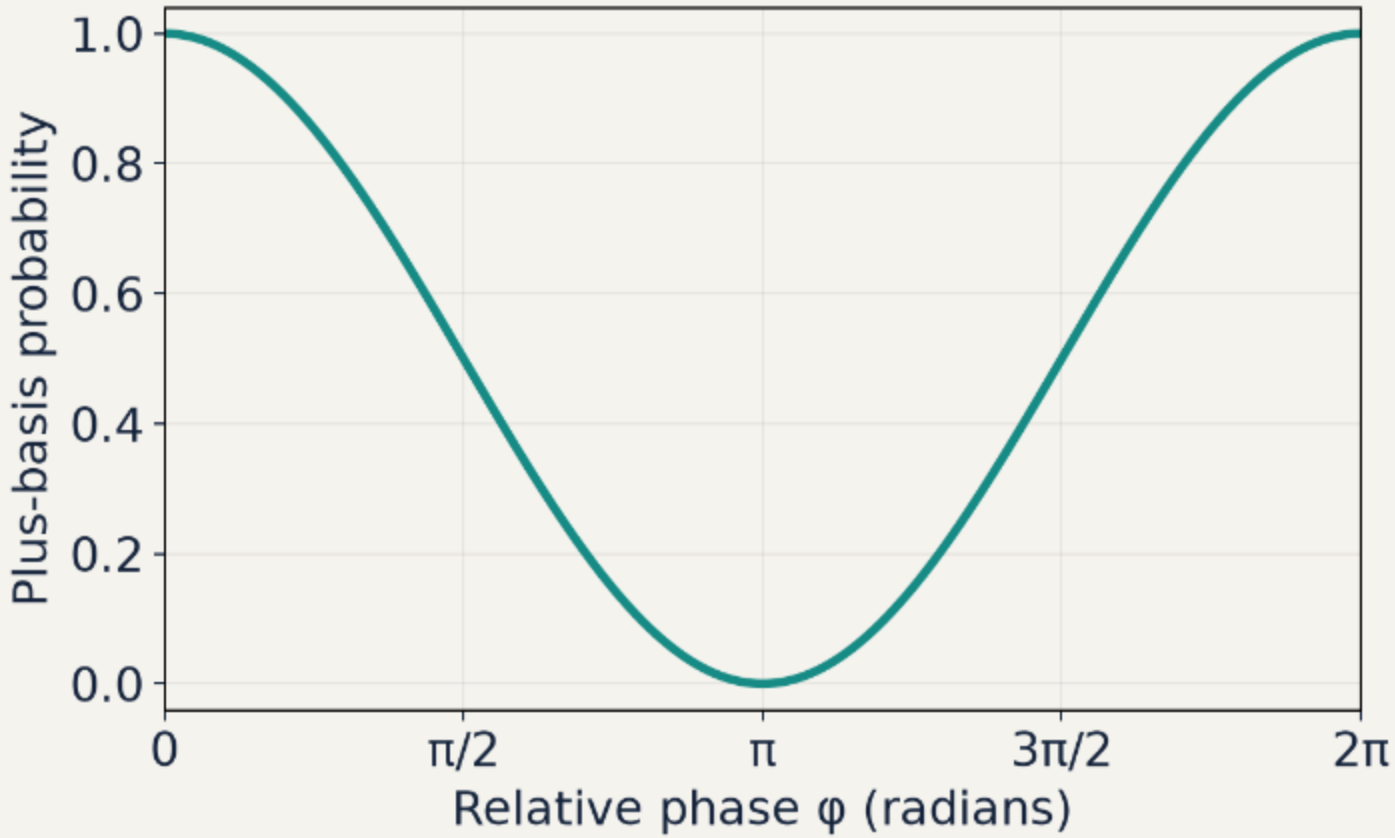
$$\alpha' = e^{(i\pi/2)}\,\alpha = i(1 + i)/2$$

$$\alpha' = (-1 + i)/2$$

$$|\alpha'|^2 = 1/2 = |\alpha|^2$$

Multiplying an amplitude by a unit phase preserves its squared magnitude.

Phase can affect an interference result



$A_+ = (1 + e^{(i\varphi)})/2$

$p_+ = |A_+|^2 = (1 + \cos\varphi)/2$

At $\varphi = 0$: $p_+ = 1$

At $\varphi = \pi$: $p_+ = 0$

Amplitudes: $1/\sqrt{2}$ and $e^{(i\varphi)}/\sqrt{2}$. Measurement: the plus/minus basis.

Code walkthrough 1 · NumPy arithmetic

```
import numpy as np
z = 3 + 4j
print(z.real, z.imag)           # 3.0 4.0
print(np.conjugate(z))          # (3-4j)
print(abs(z), abs(z)**2)        # 5.0 25.0
print(np.angle(z))              # 0.927295...
assert np.isclose(abs(z)**2, 25.0)
```

Python uses ** for powers. NumPy angles use radians.

Code walkthrough 2 · QuTiP correspondence

```
import numpy as np
import qutip as qt
a = (1 + 1j) / 2
b = 1 / np.sqrt(2)
psi = a*qt.basis(2, 0) + b*qt.basis(2, 1)
p0 = qt.expect(qt.basis(2, 0).proj(), psi)
assert np.isclose(p0, abs(a)**2)
```

For this normalized state, both calculations give $p_o = 1/2$.

Practice · Calculate before running code

1. For $z = -3 + 4i$, find z^*, $|z|^2$ and its quadrant.

2. For $\alpha = (1 - i)/2$, compare α^2 with $|\alpha|^2$.

3. For $z = 2\,e^{(i\pi/3)}$, find its Cartesian coordinates.

4. Complete starter.py and compare with solution.py.

Exit check and the next lesson

Why does conjugation make $|z|^2$ real and nonnegative?

Why does multiplying by e^(iφ) preserve a magnitude?

How can relative phase change an interference probability?

Next: S02 develops vectors, inner products and matrices.

Appendix · Worked practice answers

$(-3 + 4i)^* = -3 - 4i$ $\quad$ $|-3 + 4i|^2 = 25$
$((1 - i)/2)^2 = -i/2$ $\quad$ $|(1 - i)/2|^2 = 1/2$
$2\,e^{(i\pi/3)} = 2(1/2 + i\sqrt{3}/2) = 1 + i\sqrt{3}$

Quadrant II gives atan2(4, −3) ≈ 2.214297 rad.

The phase of a nonzero number is equivalent modulo 2π.

Appendix · Global and relative phase

ψ = (α, β)
Global phase: e^(iχ)ψ = (e^(iχ)α, e^(iχ)β)
Relative phase change: (α, e^(iφ)β)

A common phase cancels from every measurement probability.

A relative phase can change probabilities in another measurement basis.

Appendix · Why Euler’s formula works

$e^{(i\varphi)} = 1 + i\varphi + (i\varphi)^2/2! + (i\varphi)^3/3! + \cdots$
Real terms: $1 - \varphi^2/2! + \varphi^4/4! - \cdots = \cos\varphi$
Imaginary terms: $i(\varphi - \varphi^3/3! + \cdots) = i \sin\varphi$

Powers of i repeat: 1, i, −1, −i, then 1 again.

Appendix · Running the supplied files

```
Windows · PowerShell (from the repository root)
.\.venv\Scripts\python.exe lectures/S01-complex-numbers/examples.py
.\.venv\Scripts\python.exe lectures/S01-complex-numbers/starter.py

Ubuntu · Bash (from the repository root)
./.venv/bin/python lectures/S01-complex-numbers/examples.py
./.venv/bin/python lectures/S01-complex-numbers/starter.py
```

Use solution.py in place of starter.py to check the reference.